$$
\begin{aligned}
&C_{lu}{}^{i1\_i2\_i3\_i4\_} \to \\
&\frac{1}{16\pi^2}\Big(-\frac{4}{81}\sum_p g_1{}^4\, LF_{3,0}[m_{\tilde{d}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{54}\sum_p g_1{}^4\, LF_{4,-1}[m_{\tilde{d}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{16}{405}\sum_p g_1{}^4\, LF_{5,-2}[m_{\tilde{d}}{}^p] \\
&\quad\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{27}\sum_p g_1{}^4\, LF_{3,0}[m_{\tilde{e}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{18}\sum_p g_1{}^4\, LF_{4,-1}[m_{\tilde{e}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{16}{135}\sum_p g_1{}^4\, LF_{5,-2}[m_{\tilde{e}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}\sum_p g_1{}^4\, LF_{3,0}[m_{\tilde{l}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{5}{36}\sum_p g_1{}^4\, LF_{4,-1}[m_{\tilde{l}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{135}\sum_p g_1{}^4\, LF_{5,-2}[m_{\tilde{l}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{2}{81}\sum_p g_1{}^4\, LF_{3,0}[m_{\tilde{q}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{108}\sum_p g_1{}^4\, LF_{4,-1}[m_{\tilde{q}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{8}{405}\sum_p g_1{}^4\, LF_{5,-2}[m_{\tilde{q}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{16}{81}\sum_p g_1{}^4\, LF_{3,0}[m_{\tilde{u}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{10}{27}\sum_p g_1{}^4\, LF_{4,-1}[m_{\tilde{u}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{64}{405}\sum_p g_1{}^4\, LF_{5,-2}[m_{\tilde{u}}{}^p]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{2}{27}g_1{}^4\, LF_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{9}g_1{}^4\, LF_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{16}{135}g_1{}^4\, LF_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{72}g_1{}^4\, LF_{2,1,0}[m_1, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{72}g_1{}^4\, LF_{2,2,-1}[m_1, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{1}{36}g_1{}^4\, LF_{3,1,-1}[m_1, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{72}g_1{}^4\, LF_{4,1,-2}[m_1, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{72}g_1{}^4\, LF_{2,1,0}[m_1, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{72}g_1{}^4\, LF_{2,2,-1}[m_1, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{1}{36}g_1{}^4\, LF_{3,1,-1}[m_1, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{72}g_1{}^4\, LF_{4,1,-2}[m_1, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{2}{81}g_1{}^4\, LF_{2,1,0}[m_1, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}g_1{}^4\, LF_{2,2,-1}[m_1, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{4}{81}g_1{}^4\, LF_{3,1,-1}[m_1, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}g_1{}^4\, LF_{4,1,-2}[m_1, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{2}{81}g_1{}^4\, LF_{2,1,0}[m_1, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}g_1{}^4\, LF_{2,2,-1}[m_1, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{4}{81}g_1{}^4\, LF_{3,1,-1}[m_1, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}g_1{}^4\, LF_{4,1,-2}[m_1, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{2,1,0}[m_2, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{2,2,-1}[m_2, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{1}{12}g_1{}^2\, g_2{}^2\, LF_{3,1,-1}[m_2, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{4,1,-2}[m_2, m_{\tilde{l}}{}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{2,1,0}[m_2, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{2,2,-1}[m_2, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{1}{12}g_1{}^2\, g_2{}^2\, LF_{3,1,-1}[m_2, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{4,1,-2}[m_2, m_{\tilde{l}}{}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{2,1,0}[m_3, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{2,2,-1}[m_3, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{4}{27}g_1{}^2\, g_3{}^2\, LF_{3,1,-1}[m_3, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{4,1,-2}[m_3, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{2,1,0}[m_3, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{2,2,-1}[m_3, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{4}{27}g_1{}^2\, g_3{}^2\, LF_{3,1,-1}[m_3, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{4,1,-2}[m_3, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{2}{9}g_1{}^2\,\overline{y_e}{}^{i2p}\,y_e{}^{i1p}\, LF_{2,1,0}[m_{\tilde{e}}{}^p, \tilde{\mu}]\,\delta_{i3i4}-\frac{1}{9}g_1{}^2\,\overline{y_e}{}^{i2p}\,y_e{}^{i1p}\, LF_{2,2,-1}[m_{\tilde{e}}{}^p, \tilde{\mu}]\,\delta_{i3i4}- \\
&\quad\frac{1}{9}g_1{}^2\,\overline{y_e}{}^{i2p}\,y_e{}^{i1p}\, LF_{3,1,-1}[m_{\tilde{e}}{}^p, \tilde{\mu}]\,\delta_{i3i4}+\frac{1}{36}g_1{}^4\, LF_{2,1,0}[m_{\tilde{l}}{}^{i1}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{72}g_1{}^4\, LF_{3,1,-1}[m_{\tilde{l}}{}^{i1}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{12}g_1{}^2\, g_2{}^2\, LF_{2,1,0}[m_{\tilde{l}}{}^{i1}, m_2]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{3,1,-1}[m_{\tilde{l}}{}^{i1}, m_2]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{36}g_1{}^4\, LF_{2,1,0}[m_{\tilde{l}}{}^{i2}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{72}g_1{}^4\, LF_{3,1,-1}[m_{\tilde{l}}{}^{i2}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{12}g_1{}^2\, g_2{}^2\, LF_{2,1,0}[m_{\tilde{l}}{}^{i2}, m_2]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{24}g_1{}^2\, g_2{}^2\, LF_{3,1,-1}[m_{\tilde{l}}{}^{i2}, m_2]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}g_1{}^2\,\overline{y_u}{}^{pi3}\,y_u{}^{pi4}\, LF_{2,1,0}[m_{\tilde{q}}{}^p, \tilde{\mu}]\,\delta_{i1i2}- \\
&\quad\frac{1}{36}g_1{}^2\,\overline{y_u}{}^{pi3}\,y_u{}^{pi4}\, LF_{2,2,-1}[m_{\tilde{q}}{}^p, \tilde{\mu}]\,\delta_{i1i2}-\frac{1}{36}g_1{}^2\,\overline{y_u}{}^{pi3}\,y_u{}^{pi4}\, LF_{3,1,-1}[m_{\tilde{q}}{}^p, \tilde{\mu}]\,\delta_{i1i2}+ \\
&\quad\frac{4}{81}g_1{}^4\, LF_{2,1,0}[m_{\tilde{u}}{}^{i3}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}g_1{}^4\, LF_{3,1,-1}[m_{\tilde{u}}{}^{i3}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{4}{27}g_1{}^2\, g_3{}^2\, LF_{2,1,0}[m_{\tilde{u}}{}^{i3}, m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{3,1,-1}[m_{\tilde{u}}{}^{i3}, m_3]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{4}{81}g_1{}^4\, LF_{2,1,0}[m_{\tilde{u}}{}^{i4}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{81}g_1{}^4\, LF_{3,1,-1}[m_{\tilde{u}}{}^{i4}, m_1]\,\delta_{i1i2}\,\delta_{i3i4}+ \\
&\quad\frac{4}{27}g_1{}^2\, g_3{}^2\, LF_{2,1,0}[m_{\tilde{u}}{}^{i4}, m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{27}g_1{}^2\, g_3{}^2\, LF_{3,1,-1}[m_{\tilde{u}}{}^{i4}, m_3]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{9}g_1{}^2\,\overline{y_e}{}^{i2p}\,y_e{}^{i1p}\, LF_{2,1,0}[\tilde{\mu}, m_{\tilde{e}}{}^p]\,\delta_{i3i4}+\frac{1}{18}g_1{}^2\,\overline{y_e}{}^{i2p}\,y_e{}^{i1p}\, LF_{3,1,-1}[\tilde{\mu}, m_{\tilde{e}}{}^p]\,\delta_{i3i4}- \\
&\quad\frac{1}{18}g_1{}^2\,\overline{y_e}{}^{i2p}\,y_e{}^{i1p}\, LF_{4,1,-2}[\tilde{\mu}, m_{\tilde{e}}{}^p]\,\delta_{i3i4}-\frac{1}{36}g_1{}^2\,\overline{y_u}{}^{pi3}\,y_u{}^{pi4}\, LF_{2,1,0}[\tilde{\mu}, m_{\tilde{q}}{}^p]\,\delta_{i1i2}+ \\
&\quad\frac{11}{36}g_1{}^2\,\overline{y_u}{}^{pi3}\,y_u{}^{pi4}\, LF_{3,1,-1}[\tilde{\mu}, m_{\tilde{q}}{}^p]\,\delta_{i1i2}-\frac{1}{9}g_1{}^2\,\overline{y_u}{}^{pi3}\,y_u{}^{pi4}\, LF_{4,1,-2}[\tilde{\mu}, m_{\tilde{q}}{}^p]\,\delta_{i1i2}- \\
&\quad\frac{1}{18}g_1{}^4\, LF_{2,1,1,-1}[m_1, m_{\tilde{l}}{}^{i1}, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{9}g_1{}^4\, m_1{}^2\, LF_{2,1,1,0}[m_1, m_{\tilde{l}}{}^{i1}, m_{\tilde{u}}{}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{18}g_1{}^4\, LF_{2,1,1,-1}[m_1, m_{\tilde{l}}{}^{i2}, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{9}g_1{}^4\, m_1{}^2\, LF_{2,1,1,0}[m_1, m_{\tilde{l}}{}^{i2}, m_{\tilde{u}}{}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}- \\
&\quad\frac{1}{2}\tilde{\mu}^2\,\overline{y_e}{}^{i2p}\,y_e{}^{i1p}\,\overline{y_u}{}^{ri3}\,y_u{}^{ri4}\, LF_{2,1,1,0}[\tilde{\mu}, m_{\tilde{e}}{}^p, m_{\tilde{q}}{}^r]\Big)
\end{aligned}
$$